

Contents

1	Log setups	1
1.1	Namespace log N1 - Simple namespace creation and deletion	1
1.2	Namespace log N2 - Namespace with resource quotas	1
1.3	Pod log P1 - Simple Pod creation and inspection	2
1.4	Pod log P2 - Image download	3
1.5	ConfigMap C1 - Create a ConfigMap and use it in a Pod, then delete it	4
1.6	Secret S1 - Create a Secret and use it in a Pod, then delete it	5
1.7	Node log N01 - Node inspection	6
1.8	Certificate log CSR1 - Create a CertificateSigningRequest and approve it	6
1.9	Role log R1 - Create a Role and bind it to a User	8
1.10	Role log R2 - Create a ClusterRole and bind it to a Group	9
1.11	Role log R3 - Create a ServiceAccount for a long-running Pod and bind it to a Role . .	10
1.12	Access Review AR1 - Use can-i to view permissions of self, other users, over different namespaces	11
1.13	TODO Access Review AR2 - Use auth whoami to view the current user	14
1.14	TODO Access Review AR3 - Impersonating a user	15

1 Log setups

1.1 Namespace log N1 - Simple namespace creation and deletion

- Create a namespace:

```
kubectl create namespace rising
```

- Get the namespace:

```
kubectl get namespaces
```

- Describe the namespace:

```
kubectl describe namespace rising
```

- Delete the namespace:

```
kubectl delete namespace rising
```

1.2 Namespace log N2 - Namespace with resource quotas

- **Prerequisite:** recreate the `rising` namespace:

```
kubectl create namespace rising
```

- Create a ResourceQuota for the namespace:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: ResourceQuota
  metadata:
    name: rising-quota
  spec:
    hard:
      pods: "1"
      requests.cpu: "1"
      requests.memory: 1Gi
      limits.cpu: "2"
      limits.memory: 2Gi
EOF
```

- Inspect the ResourceQuota:

```
kubectl -n rising describe resourcequota rising-quota
```

- Create a Pod that will exceed the quota:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: over-quota-pod
    namespace: rising
  spec:
    containers:
      - name: over-quota-container
        image: nginx
        resources:
          requests:
            cpu: "1"
            memory: 1.5Gi
          limits:
            cpu: "2"
            memory: 2Gi
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Now create a Pod that will respect the quota:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: within-quota-pod
    namespace: rising
  spec:
    containers:
      - name: within-quota-container
        image: nginx
        resources:
          requests:
            cpu: "1"
            memory: 1Gi
          limits:
            cpu: "2"
            memory: 2Gi
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Run a further Pod, it will fail:

```
kubectl -n rising run --image=nginx nginx
```

- Clean everything up, one pod will fail because it was never created:

```
kubectl -n rising delete pod over-quota-pod
kubectl -n rising delete pod within-quota-pod
kubectl delete resourcequota rising-quota
```

1.3 Pod log P1 - Simple Pod creation and inspection

- Create a pod:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: rising-pod
    namespace: rising
  spec:
    containers:
      - name: first-container
        image: nginx
      - name: second-container
        image: alpine
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Inspect the Pod:

```
kubectl -n rising describe pod rising-pod
```

- Patch the Pod and try to add a third container, this will fail:

```
kubectl -n rising patch pod rising-pod --type='json' -p='[{"op": "add", "path":
↳ "/spec/containers/1", "value": {"name": "third-container", "image": "nginx"}}]'
```

- Get the pod log:

```
kubectl -n rising logs rising-pod -c second-container
```

- Execute commands in the second containers, the first will complain about the shell, the second will work:

```
kubectl -n rising exec -t rising-pod -c second-container -- /bin/bash -c "echo Hello again,
↳ Kubernetes!"
kubectl -n rising exec -t rising-pod -c second-container -- /bin/sh -c "echo Hello again,
↳ Kubernetes!"
```

- Delete the pod:

```
kubectl -n rising delete pod rising-pod
```

1.4 Pod log P2 - Image download

- **Prerequisite:** run the following helper script to wipe the image cache on all nodes:

```
for node in $(kubectl get nodes -o jsonpath='{.items[*].metadata.name}'); do
  echo "Pruning images on $node"
  ssh -F ssh/cluster-setup $node "sudo crictl rmi --prune"
done
```

- Create a Pod with a container that uses a non-existent image:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: image-pod
    namespace: rising
  spec:
    containers:
      - name: image-container
        image: www.fbk.eu/non-existent-image
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Get the pod log:

```
kubectl -n rising logs image-pod -c image-container
```

- Get the events that led to the failure:

```
kubectl get events -n rising --sort-by='.metadata.creationTimestamp' --selector
↳ 'involvedObject.name=image-pod'
```

- Delete the pod:

```
kubectl -n rising delete pod image-pod
```

- Create a pod with an image that will surely not be in cache:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: image-pod
    namespace: rising
  spec:
    containers:
      - name: image-container
        image: busybox
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Wait for the Pod to be downloaded and monitor the events:

```
kubectl wait --for=condition=Ready pod/image-pod -n rising
kubectl get events -n rising --sort-by='.metadata.creationTimestamp'
```

- Delete the pod:

```
kubectl -n rising delete pod image-pod
```

1.5 ConfigMap C1 - Create a ConfigMap and use it in a Pod, then delete it

- Create a ConfigMap:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: ConfigMap
  metadata:
    name: my-config
    namespace: rising
  data:
    status: "decent"
EOF
```

- Get the configmap

```
kubectl -n rising get configmap my-config -o yaml
```

- Deploy a Pod that uses the ConfigMap:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: config-pod
    namespace: rising
  spec:
    containers:
      - name: config-container
        image: alpine
        command: ["sh", "-c", 'echo Status is \${STATUS} && sleep 3600']
        env:
          - name: STATUS
            valueFrom:
              configMapKeyRef:
                name: my-config
                key: status
EOF
```

- Access the logs of the Pod:

```
kubectl wait --for=condition=Ready pod/config-pod -n rising
kubectl -n rising logs config-pod -c config-container
```

- Clean everything up:

```
kubectl -n rising delete pod config-pod
kubectl -n rising delete configmap my-config
```

1.6 Secret S1 - Create a Secret and use it in a Pod, then delete it

- Create a secret

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Secret
  metadata:
    name: my-secret
    namespace: rising
  type: Opaque
  data:
    username: $(echo -n "admin" | base64)
    password: $(echo -n "password" | base64)
EOF
```

- Inspect the secret:

```
kubectl -n rising get secret my-secret -o yaml
```

- Attach the secret to a Pod:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: secret-pod
    namespace: rising
  spec:
    containers:
      - name: secret-container
        image: alpine
        command: ["sh", "-c", 'echo Username is \${USERNAME} && echo Password is \${PASSWORD} && sleep 3600']
        env:
          - name: USERNAME
            valueFrom:
              secretKeyRef:
                name: my-secret
                key: username
          - name: PASSWORD
            valueFrom:
              secretKeyRef:
                name: my-secret
                key: password
EOF
```

- Inspect the Pod:

```
kubectl -n rising describe pod secret-pod
```

- Access the logs of the Pod:

```
kubectl wait --for=condition=Ready pod/secret-pod -n rising
kubectl -n rising logs secret-pod -c secret-container
```

- Clean everything up:

```
kubectl -n rising delete pod secret-pod
kubectl -n rising delete secret my-secret
```

1.7 Node log N01 - Node inspection

- Assess the available nodes in the cluster:

```
kubectl get nodes
```

- Inspect one of the nodes:

```
kubectl describe node kubeadm-worker1
```

1.8 Certificate log CSR1 - Create a CertificateSigningRequest and approve it

- Note: an helper script is available in `scripts/generate_k8s_user.sh` that automates what is done in this section.
- Use openssl to generate a key and a certificate signing request:

```
export TMPDIR=$(mktemp -d)
export ORIGINDIR=$(pwd)
cd $TMPDIR
export KUBEUSER=malicious-user
openssl genpkey -out ${KUBEUSER}.key -algorithm Ed25519
openssl req -new -key ${KUBEUSER}.key -out ${KUBEUSER}.csr -subj
↳ "/CN=${KUBEUSER}/O=rising/O=fbk/"
echo "Key created, available in ${TMPDIR} for user ${KUBEUSER}"
```

- Create a CertificateSigningRequest:

```
cat <<EOF | kubectl apply -f -
  apiVersion: certificates.k8s.io/v1
  kind: CertificateSigningRequest
  metadata:
    name: ${KUBEUSER}
  spec:
    request: $(cat ${KUBEUSER}.csr | base64 | tr -d '\n')
    signerName: kubernetes.io/kube-apiserver-client
    usages:
      - client auth
  EOF
```

- Deny the CertificateSigningRequest:

```
kubectl certificate deny ${KUBEUSER}
```

- Delete the failed CSR and generate a new keypair for a legitimate user:

```
rm ${KUBEUSER}.csr ${KUBEUSER}.key
kubectl delete csr ${KUBEUSER}
export KUBEUSER=rising-user
openssl genpkey -out ${KUBEUSER}.key -algorithm Ed25519
openssl req -new -key ${KUBEUSER}.key -out ${KUBEUSER}.csr -subj
↳ "/CN=${KUBEUSER}/O=rising/O=fbk/"
echo "Key created, available in ${TMPDIR} for user ${KUBEUSER}"
```

- Generate a new CertificateSigningRequest:

```
cat <<EOF | kubectl apply -f -
  apiVersion: certificates.k8s.io/v1
  kind: CertificateSigningRequest
  metadata:
    name: ${KUBEUSER}
  spec:
    request: $(cat ${KUBEUSER}.csr | base64 | tr -d '\n')
    signerName: kubernetes.io/kube-apiserver-client
    usages:
      - client auth
  EOF
```

- Inspect the status of the CSR:

```
kubectl get csr ${KUBEUSER} -o yaml
```

- Approve the CertificateSigningRequest:

```
kubectl certificate approve ${KUBEUSER}
```

- Obtain the certificate from the CSR and store it in the folder

```
kubectl get csr ${KUBEUSER} -o jsonpath='{.status.certificate}' | base64 -d > ${KUBEUSER}.cert
head -n 2 ${KUBEUSER}.cert
```

- Use the certificate to generate a .kubeconfig for the user:

```
cat <<EOF > ${KUBEUSER}.kubeconfig
  apiVersion: v1
  kind: Config
  preferences: {}
  clusters:
    $(kubectl config view --raw --minify --flatten | jq .clusters)
  contexts: []
  users: []
EOF
kubectl --kubeconfig ${KUBEUSER}.kubeconfig config set-credentials ${KUBEUSER}
↪ --client-certificate=${KUBEUSER}.cert --client-key=${KUBEUSER}.key --embed-certs=true
kubectl --kubeconfig ${KUBEUSER}.kubeconfig config set-context ${KUBEUSER}
↪ --cluster=kubernetes --user=${KUBEUSER} --cluster=risinglab
kubectl --kubeconfig ${KUBEUSER}.kubeconfig config use-context ${KUBEUSER}
cp ${KUBEUSER}.kubeconfig ${ORIGINDIR}
```

1.9 Role log R1 - Create a Role and bind it to a User

- **Prerequisite:** complete CSR1 and obtain the credentials for the user. Make sure they are available as rising-user.kubeconfig in the current directory.
- Create a Role:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    name: rising-reader
    namespace: rising
  rules:
  - apiGroups: [""]
    resources: ["pods"]
    verbs: ["get", "list"]
EOF
```

- List the roles in the namespace:

```
kubectl -n rising get roles
```

- We changed idea: let's also give get, list, patch and watch for Secrets:

```
kubectl patch role rising-reader -n rising --type=json' -p='[{"op": "add", "path":
↪ "/rules/0", "value": {"apiGroups": [""], "resources": ["secrets"], "verbs": ["get",
↪ "list", "patch", "watch"]}]']
```

- Let's create a RoleBinding and assign it to the user:


```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: RoleBinding
  metadata:
    name: pod-reader-binding
    namespace: rising
  subjects:
  - kind: User
    name: rising-user
  roleRef:
    kind: Role
    name: rising-reader
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Test the permissions of the user:

```
kubectl --kubeconfig rising-user.kubeconfig get pods -n rising
kubectl --kubeconfig rising-user.kubeconfig get secrets -n rising
kubectl --kubeconfig rising-user.kubeconfig delete pod non-existent-pod -n rising
```

1.10 Role log R2 - Create a ClusterRole and bind it to a Group

- **Prerequisite:** complete CSR1 and obtain the credentials for the user. Make sure they are available as `rising-user.kubeconfig` in the current directory.
- **Prerequisite:** complete R1 and keep the `rising-reader` Role.
- Create a ClusterRole:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRole
  metadata:
    name: cluster-reader
  rules:
  - apiGroups: [""]
    resources: ["pods"]
    verbs: ["get", "list"]
  - apiGroups: [""]
    resources: ["secrets"]
    verbs: ["get", "list", "patch", "watch"]
EOF
```

- Actually, let's straight out replace the ClusterRole with a new one:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRole
  metadata:
    name: cluster-reader
  rules:
  - apiGroups: [""]
    resources: ["nodes"]
    verbs: ["get", "list"]
EOF
```

- Create a ClusterRoleBinding and assign it to the `fbk` group:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRoleBinding
  metadata:
    name: cluster-reader-binding
  subjects:
    - kind: Group
      name: fbk
  roleRef:
    kind: ClusterRole
    name: cluster-reader
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Test the permissions of the user:

```
kubectl --kubeconfig rising-user.kubeconfig get nodes
kubectl --kubeconfig rising-user.kubeconfig get pods -n rising
kubectl --kubeconfig rising-user.kubeconfig get configmaps -n rising
kubectl --kubeconfig rising-user.kubeconfig get pods -n kube-system
```

- Clean everything up:

```
kubectl delete rolebinding pod-reader-binding -n rising
kubectl delete role rising-reader -n rising
kubectl delete clusterrolebinding cluster-reader-binding
kubectl delete clusterrole cluster-reader
kubectl delete csr rising-user
```

1.11 Role log R3 - Create a ServiceAccount for a long-running Pod and bind it to a Role

- Set up a ServiceAccount:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: ServiceAccount
  metadata:
    name: long-running-sa
    namespace: rising
EOF
```

- Create a Role:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    name: long-running-role
    namespace: rising
  rules:
    - apiGroups: [""]
      resources: ["pods"]
      verbs: ["get", "list"]
EOF
```

- Bind the ServiceAccount to the Role:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: RoleBinding
  metadata:
    name: long-running-role-binding
    namespace: rising
  subjects:
  - kind: ServiceAccount
    name: long-running-sa
    namespace: rising
  roleRef:
    kind: Role
    name: long-running-role
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Create a Pod that uses the ServiceAccount:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: long-running-pod
    namespace: rising
  spec:
    serviceAccountName: long-running-sa
    containers:
    - name: long-running-container
      image: ubuntu
      command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Inspect the Pod:

```
kubectl -n rising describe pod long-running-pod | head -n 10
```

- Try to use the ServiceAccount from inside the Pod:

```
kubectl wait --for=condition=Ready pod/long-running-pod -n rising
kubectl -n rising exec long-running-pod -- sh -c 'apt-get -qq update; apt-get install -yqq
  curl yq >/dev/null;
  export KUBE_TOKEN=$(cat /var/run/secrets/kubernetes.io/serviceaccount/token);
  curl -sSk -H "Authorization: Bearer $KUBE_TOKEN"
  "https://$KUBERNETES_SERVICE_HOST:$KUBERNETES_PORT_443_TCP_PORT/apis/authentication.k8s.io/v1/selfsubject
  > temp.json; jq .message temp.json
  curl -sSk -H "Authorization: Bearer $KUBE_TOKEN"
  "https://$KUBERNETES_SERVICE_HOST:$KUBERNETES_PORT_443_TCP_PORT/api/v1/namespaces/rising/pods"
  > temp.json; head -n 5 temp.json'
```

- Clean everything up:

```
kubectl -n rising delete pod long-running-pod
kubectl -n rising delete role long-running-role
kubectl -n rising delete rolebinding long-running-role-binding
kubectl -n rising delete serviceaccount long-running-sa
```

1.12 Access Review AR1 - Use can-i to view permissions of self, other users, over different namespaces

- Create a set of ServiceAccounts for the logs (while this should not be the case in a real-world scenario, it is done since SAs and Users are equivalent once authorized):

```
kubectl apply -f - <<EOF
  apiVersion: v1
  kind: ServiceAccount
  metadata:
    name: alice
    namespace: rising
  ---
  apiVersion: v1
  kind: ServiceAccount
  metadata:
    name: bob
    namespace: rising
  ---
  apiVersion: v1
  kind: ServiceAccount
  metadata:
    name: charlie
    namespace: rising
EOF
```

- Create tokens for the ServiceAccounts:

```
kubectl apply -f - <<EOF
  apiVersion: v1
  kind: Secret
  metadata:
    name: alice-token
    namespace: rising
    annotations:
      kubernetes.io/service-account.name: alice
  type: kubernetes.io/service-account-token
  ---
  apiVersion: v1
  kind: Secret
  metadata:
    name: bob-token
    namespace: rising
    annotations:
      kubernetes.io/service-account.name: bob
  type: kubernetes.io/service-account-token
  ---
  apiVersion: v1
  kind: Secret
  metadata:
    name: charlie-token
    namespace: rising
    annotations:
      kubernetes.io/service-account.name: charlie
  type: kubernetes.io/service-account-token
  ---
EOF
```

- Create two Roles, one for reading Pods and one for reading Secrets:

```
kubectl apply -f - <<EOF
  apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    name: pod-reader
    namespace: rising
  rules:
  - apiGroups: [""]
    resources: ["pods"]
    verbs: ["get", "list"]
  ---
  apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    name: secret-reader
    namespace: rising
  rules:
  - apiGroups: [""]
    resources: ["secrets"]
    verbs: ["get", "list"]
EOF
```

- Bind the ServiceAccounts to the Roles:

```
kubectl apply -f - <<EOF
  apiVersion: rbac.authorization.k8s.io/v1
  kind: RoleBinding
  metadata:
    name: pod-reader-binding
    namespace: rising
  subjects:
  - kind: ServiceAccount
    name: alice
    namespace: rising
  - kind: ServiceAccount
    name: bob
    namespace: rising
  roleRef:
    kind: Role
    name: pod-reader
    apiGroup: rbac.authorization.k8s.io
  ---
  apiVersion: rbac.authorization.k8s.io/v1
  kind: RoleBinding
  metadata:
    name: secret-reader-binding
    namespace: rising
  subjects:
  - kind: ServiceAccount
    name: charlie
    namespace: rising
  roleRef:
    kind: Role
    name: secret-reader
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Obtain credentials for the ServiceAccounts and generate a kubeconfig for them:

```
kubectl get secret alice-token -n rising -o jsonpath='{.data.token}' | base64 -d > alice.token
kubectl get secret bob-token -n rising -o jsonpath='{.data.token}' | base64 -d > bob.token
kubectl get secret charlie-token -n rising -o jsonpath='{.data.token}' | base64 -d >
↵ charlie.token
```

```

cat <<EOF > alice.kubeconfig
  apiVersion: v1
  kind: Config
  preferences: {}
  clusters:
  $(kubectl config view --raw --minify --flatten | yq .clusters)
  contexts: []
  users: []
EOF
cp alice.kubeconfig bob.kubeconfig
cp alice.kubeconfig charlie.kubeconfig
kubectl --kubeconfig alice.kubeconfig config set-credentials alice --token=$(cat alice.token)
kubectl --kubeconfig bob.kubeconfig config set-credentials bob --token=$(cat bob.token)
kubectl --kubeconfig charlie.kubeconfig config set-credentials charlie --token=$(cat
↳ charlie.token)
kubectl --kubeconfig alice.kubeconfig config set-context alice --cluster=kubernetes
↳ --user=alice --cluster=risinglab
kubectl --kubeconfig bob.kubeconfig config set-context bob --cluster=kubernetes --user=bob
↳ --cluster=risinglab
kubectl --kubeconfig charlie.kubeconfig config set-context charlie --cluster=kubernetes
↳ --user=charlie --cluster=risinglab
kubectl --kubeconfig alice.kubeconfig config use-context alice
kubectl --kubeconfig bob.kubeconfig config use-context bob
kubectl --kubeconfig charlie.kubeconfig config use-context charlie
rm alice.token bob.token charlie.token

```

- Check as Alice if she can list Pods and Secrets in the namespace:

```

kubectl --kubeconfig alice.kubeconfig auth can-i list pods -n rising
kubectl --kubeconfig alice.kubeconfig auth can-i list secrets -n rising

```

- Check as Bob if he can list Pods in the namespaces rising and kube-system:

```

kubectl --kubeconfig bob.kubeconfig auth can-i list pods -n rising
kubectl --kubeconfig bob.kubeconfig auth can-i list pods -n kube-system

```

- Check as Charlie if he can list Secrets in the namespace:

```

kubectl --kubeconfig charlie.kubeconfig auth can-i list secrets -n rising

```

- Check as Charlie if he can list nodes:

```

kubectl --kubeconfig charlie.kubeconfig auth can-i list nodes

```

- Check what Charlie can do in the rising namespace:

```

kubectl --kubeconfig charlie.kubeconfig auth can-i --list -n rising

```

1.13 TODO Access Review AR2 - Use auth whoami to view the current user

- **Prerequisite:** complete AR1 and obtain the credentials for the users. Make sure they are available as `alice.kubeconfig`, `bob.kubeconfig` and `charlie.kubeconfig` in the current directory.
- Check as each of the users who they are:

```

kubectl --kubeconfig alice.kubeconfig auth whoami
kubectl --kubeconfig bob.kubeconfig auth whoami
kubectl --kubeconfig charlie.kubeconfig auth whoami

```

1.14 TODO Access Review AR3 - Impersonating a user

- **Prerequisite:** complete AR1 and obtain the credentials for the users. Make sure they are available as `alice.kubeconfig`, `bob.kubeconfig` and `charlie.kubeconfig` in the current directory.
- Augment Charlie's capabilities to let him impersonate ServiceAccounts:

```
kubectl apply -f - <<EOF
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRole
  metadata:
    name: impersonator
  rules:
    - apiGroups: [""]
      resources: ["serviceaccounts"]
      verbs: ["impersonate"]
      resourceNames: ["alice", "bob"]
  ---
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRoleBinding
  metadata:
    name: impersonator-binding
  subjects:
    - kind: ServiceAccount
      name: charlie
      namespace: rising
  roleRef:
    kind: ClusterRole
    name: impersonator
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Check if Charlie can impersonate any user or SA:

```
kubectl --kubeconfig charlie.kubeconfig auth can-i impersonate users
kubectl --kubeconfig charlie.kubeconfig auth can-i impersonate serviceaccounts
```

- Let Charlie see if Alice can list Pods in `rising`, and what she can do in `kube-system`:

```
kubectl --kubeconfig charlie.kubeconfig auth can-i list pods -n rising --as
↳ system:serviceaccount:rising:alice
kubectl --kubeconfig charlie.kubeconfig auth can-i --list -n kube-system --as
↳ system:serviceaccount:rising:alice
```

- As Charlie, let's impersonate Bob and create a Pod in the `rising` namespace, then list Pods:

```
kubectl --kubeconfig charlie.kubeconfig --as system:serviceaccount:rising:bob run
↳ --image=ubuntu --restart=Never --namespace=rising --command -- sleep 3600
kubectl --kubeconfig charlie.kubeconfig --as system:serviceaccount:rising:bob get pods -n
↳ rising
```

- As Charlie, let's impersonate Alice and try to list nodes:

```
kubectl --kubeconfig charlie.kubeconfig --as system:serviceaccount:rising:alice get nodes
```

- Clean everything up:

```
kubectl delete clusterrolebinding impersonator-binding
kubectl delete clusterrole impersonator
kubectl delete rolebinding pod-reader-binding -n rising
kubectl delete role pod-reader -n rising
kubectl delete rolebinding secret-reader-binding -n rising
```

```
kubectl delete role secret-reader -n rising
kubectl delete secret alice-token bob-token charlie-token -n rising
kubectl delete serviceaccount alice bob charlie -n rising
rm alice.kubeconfig bob.kubeconfig charlie.kubeconfig
```